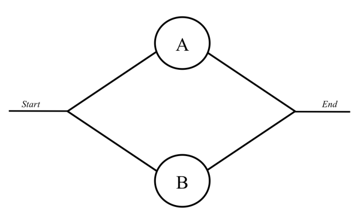
Stat 641: Section 1
White
Fall 2021
Exam 1

- This exam is closed-book and closed-note. You may not use the internet, homework, or classmates as a resource. You may not use a calculator (that includes R).
- I will not impose the four-hour limit discussed, but you must take the exam in one sitting.
- Turn in your exam to me by 9 AM on Monday (either in person or over Learning Suite).
- **Read all questions carefully**
- Simplify your answers as much as *reasonably* possible, but remember to provide a complete response. Show your work but **clearly mark your final answer** (for example, draw a box around the final answer). Please be neat and organized. Credit (including partial credit) is given for work, not only final answers.
- You may use scratch paper, but please do not turn it in.
- Please write your name on all of the pages.

1. (5 points) Let $\mathcal{F}$ a σ-algebra on some sample space S. Prove that if any $A_1, A_2, ... \in \mathcal{F}$, then $\bigcap_{n=1}^{\infty} A_n \in \mathcal{F}$ (closed under countable intersection).

2. Consider the system of two components, A and B shown below, where component B is higher quality than component A. This is a parallel system, meaning the system will function if A functions or B functions. Suppose A functions with probability p. If A functions, then B functions with probability $r > p$. If A has failed, then B functions with probability p.



   (a) (3 points) What is the probability that B functions?
   (b) (3 points) What is the probability that the system functions?
   (c) (4 points) If B does not function, what is the probability that A does?

3. Let $X \sim \text{Beta}(a, b)$. The pdf of X is

$$f(x) = \begin{cases} \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} x^{a-1}(1-x)^{b-1} & 0 < x < 1 \\ 0 & \text{otherwise,} \end{cases}$$

where $a > 0$ and $b > 0$.

(a) (5 points) Find $\mathbb{E}\left(\frac{1-X}{X}\right)$ without finding the distribution of $\frac{1-X}{X}$.

(b) (5 points) Derive the PDF $f_Y(y)$ for $Y = -\log\left(\frac{X}{1-X}\right)$, where log is the natural logarithm.

4. (6 points) Consider a random variable with pdf is

$$f(x) = \begin{cases} 2x & x \in (0,1), \\ 0 & \text{otherwise.} \end{cases}$$

Derive the CDF of $F_Y(y)$ for $Y = X(1-X)$ and prove that it is a valid CDF.

5. (5 points) Consider a random variable with pdf is

$$f(x) = \begin{cases} \frac{n(n-1)}{2^n} x^{n-2}(2-x) & x \in (0,2), \\ 0 & \text{otherwise,} \end{cases}$$

where $n \in \{1, 2, 3, ...\}$ is an unknown parameter. Using the fact that this in an exponential family, find the expected value of $\ln(X)$.

6. With $a > 0$ and $b > 0$, the pdf of a Gamma(a, b) random variable is

$$f(x) = \begin{cases} \frac{1}{\Gamma(a) b^a} x^{a-1} e^{-x/b} & x > 0, \\ 0 & \text{otherwise.} \end{cases}$$

(a) (5 points) Let $X \sim$ Gamma$(\alpha n, \beta/n)$. Find the MGF of X.

(b) (3 points) What is $\lim_{n\to\infty} M_X(t)$? (You may assume $\lim_{n\to\infty} (c_n)^d = c^d$ given that $\lim_{n\to\infty} c_n = c$) Bonus: What does this mean about the limiting distribution of X?

7. Consider a random variable X with the following triangle distribution over $\mathcal{X} = (-3, 4)$,

$$f(x) = \begin{cases} \frac{x+3}{7} & -3 < x \leq -1, \\ \frac{2(4-x)}{35} & -1 < x < 4, \\ 0 & \text{otherwise.} \end{cases}$$

(a) (4 points) What is the probability that $X > 1$ given that X is positive.

(b) (8 points) Find $f_Y(y)$ for $Y = X^2$.

8. Assume that the average length of a female great white shark is 16 ft.

(a) (2 points) Without making more assumptions, provide an upper bound on the probability that a random female great white shark is over 20 feet long.

(b) (2 points) Now suppose that the standard deviation of the length is 4/3. Now, give a lower bound on the probability that a random female great white shark is between 12 and 20 feet.